☑ CTL Provisional DECISION SHEET (PDSH)
☐ CTL DECISION SHEET (DSH)

Standard(s) (incl. year)	**Subclause(s)**	**Tracking No.**
IEC 61010-1:2010 IEC 61010-1:2010/AMD1:2016	9.1. b) 9.2. a) 1) 9.4.	2307A
Category		
MEAS		
Subject	**Keywords**	**Developed by**
Limited Energy Circuit	Circuits derived from limited energy circuits	ETF 3

Question

Where for an equipment or equipment area / subassembly, the protection against the spread of fire in clause 9 relies on 9.1. b) and the compliance path via 9.2.a) 1) is chosen:

Is a circuit derived from a limited energy circuit again subject to an evaluation to 9.4 a) and b) 1) to confirm it is still limited energy after voltage changes (such as reduction from 24Vdc to 12Vdc)?

Decision

No. Clause 9.2 a) 1) requires the evaluation of a circuit or part of the equipment to determine if it is limited as per 9.4. If the circuit is found to meet the requirements of 9.4, a further evaluation on circuits derived from this (energy limited) circuit is not necessary, as the possibility of ignition and occurrence of fire has already been judged as being reduced to a tolerable level.

Explanatory notes

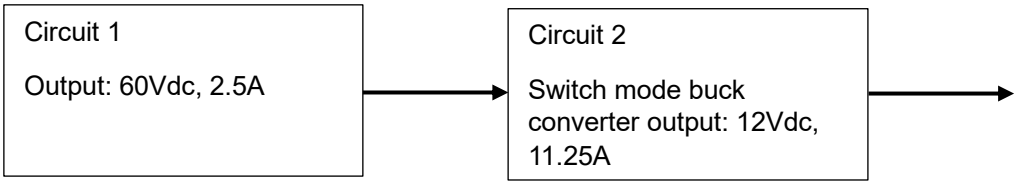
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DSH	
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The resolution was determined by ETF-3 meeting discussing the requirements. Clause 9.2 a) 1) specifically references the need to evaluate a circuit as limited per 9.2. After that determination and classification of the circuit per clause 9.4, the risk of fire has been determined on that circuit. Clause 9.4, item c) is critical to ensure that separation of basic insulation between the limited energy circuits under investigation and any other circuit is maintained. No future investigation review of the circuit is required when compliance with 9.4 is determined.

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Example 1.

Circuit 2 is a simple buck converter with 90% efficiency, and it is derived from circuit 1. Output of the circuit 1 meets the requirements of 9.4 a) and b) 1) but the output of the circuit 2 does not meet the requirements of 9.4 b) 1).



Example 2.

Circuit 2 is a simple boost converter with 90% efficiency, and it is derived from circuit 1. Output of the circuit 1 meets the requirements of 9.4 a) and b) 1) but the output of the circuit 2 does not meet the requirements of 9.4 a).

